

Dark Network Mesh & Post- Quantum Cryptography



Team Members

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Background & Security Challenges

Quantum Threat

HARVEST NOW, DECRYPT LATER

- Encryption Risk: RSA & ECC encryption may become obsolete against future quantum computers.
- Data Harvesting: Attackers steal encrypted data today to decrypt it in the future.
- Long-Term Threat: High risk for sensitive financial data.

Network Discovery

RECONNAISSANCE RISKS

- Public Exposure: Servers processing 5M daily transactions remain discoverable.
- Reconnaissance: Susceptible to unauthorized port scanning & service discovery.
- Pre-Attack Risk: Enables adversaries to gather network info before main attacks.

Legacy VPN Limits

ARCHITECTURAL RESILIENCE

- No Obfuscation: Traditional VPNs do not hide services or servers from external users.
- Hub-and-Spoke Model: Centralized architecture creates a Single Point of Failure (SPOF).
- Business Impact: Central hub failure directly disrupts B2B partner communication.

Three-Layer Security Architecture

LAYER 01

Dark Network (SPA)

Utilizes fwknop to keep VPN ports (51820/UDP) closed by default. Services remain invisible to network reconnaissance until a valid Single Packet Authorization (SPA) payload is authenticated.

MECHANISM: FWKNOP

LAYER 02

Transport Encryption

WireGuard establishes high-performance mesh tunnels using Curve25519. This creates secure P2P communication channels for financial data that bypass the vulnerabilities of centralized hub-and-spoke models.

PROTOCOL: WIREGUARD

LAYER 03

Post-Quantum Layer

Rosenpass performs periodic ML-KEM-1024 key exchanges, feeding quantum-resistant pre-shared keys into the WireGuard session to protect against "Harvest Now, Decrypt Later" attacks.

ALGORITHM: ML-KEM

Architecture Diagram

A layered hub-and-spoke-over-mesh model deployed on Azure VPS, securing B2B financial transactions against modern and quantum threats.

LAYER 01

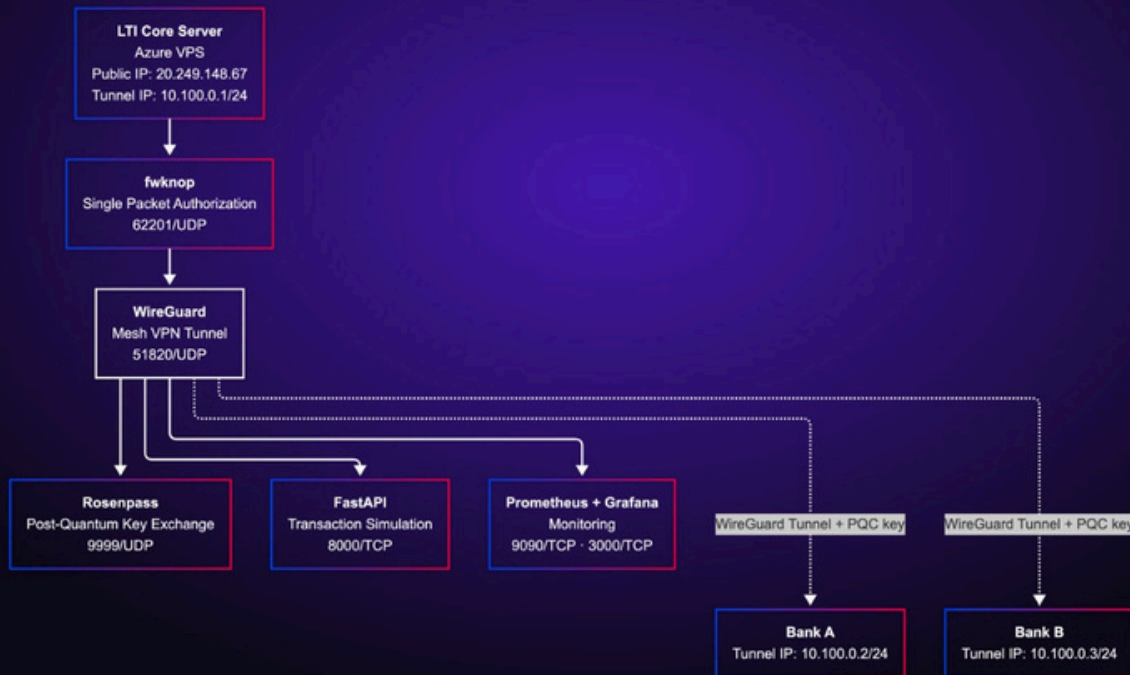
Dark Network: SPA hides the WireGuard port (51820/UDP) until a valid 62201/UDP packets is verified

LAYER 02

WireGuard Mesh: Secure transport tunnel for mutual peer authentication

LAYER 03

Rosenpass (PQC): ML-KEY-1024 Key exchange, rotates to prevent “Harvest now, Decrypt Later.”



Handshake & Authorization Workflow

1

Reconnaissance Blocked

Client scans port 51820 — server returns filtered/hidden. No service is detectable.

2

Single Packet Authorization

Client sends encrypted + HMAC-signed SPA packet; fwknop opens the port for a 10s window.

3

WireGuard Handshake

Client initiates Curve25519 handshake; encrypted tunnel established, port closes again.

4

PQC Key Exchange (every 2 min)

Rosenpass performs ML-KEM-1024 handshake in parallel, injecting fresh PSK into tunnel.

PARTICIPANTS IN THE FLOW

Client (Bank)

Initiates SPA, WG handshake, and app requests.

fwknop (62201/UDP)

Validates SPA; opens/closes WG port.

WireGuard (51820/UDP)

Establishes encrypted transport tunnel.

Rosenpass (9999/UDP)

ML-KEM-1024 exchange for session PSK.

FastAPI (8000/TCP)

Responds to transaction requests in-tunnel.

Application-Layer Verification

Besides proving that the tunnel is secure, the system also needs to show that it can carry real application traffic, similar to a real bank connection. A dummy transaction API was created using FastAPI and deployed on the LTI Server. The API is bound only to the internal tunnel interface (10.100.0.1:8000), so it can only be accessed after a bank node successfully completes the WireGuard handshake and establishes the VPN tunnel. Any request made from outside the tunnel will fail because the API is not available on the public network.

REQUEST (FROM INSIDE THE TUNNEL)

```
curl -X POST http://10.100.0.1:8000/v1/transaction \
-H "Content-Type: application/json" \
-d '{"amount": 100000, "recipient": "Bank B"}
```

RESPONSE

```
{
  "tx_id": "TX-1720850745123",
  "status": "success",
  "amount": 100000,
  "recipient": "Bank B"
}
```

DEMONSTRATION

DNM-PQC-VPN GROUP 3

THANK YOU

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